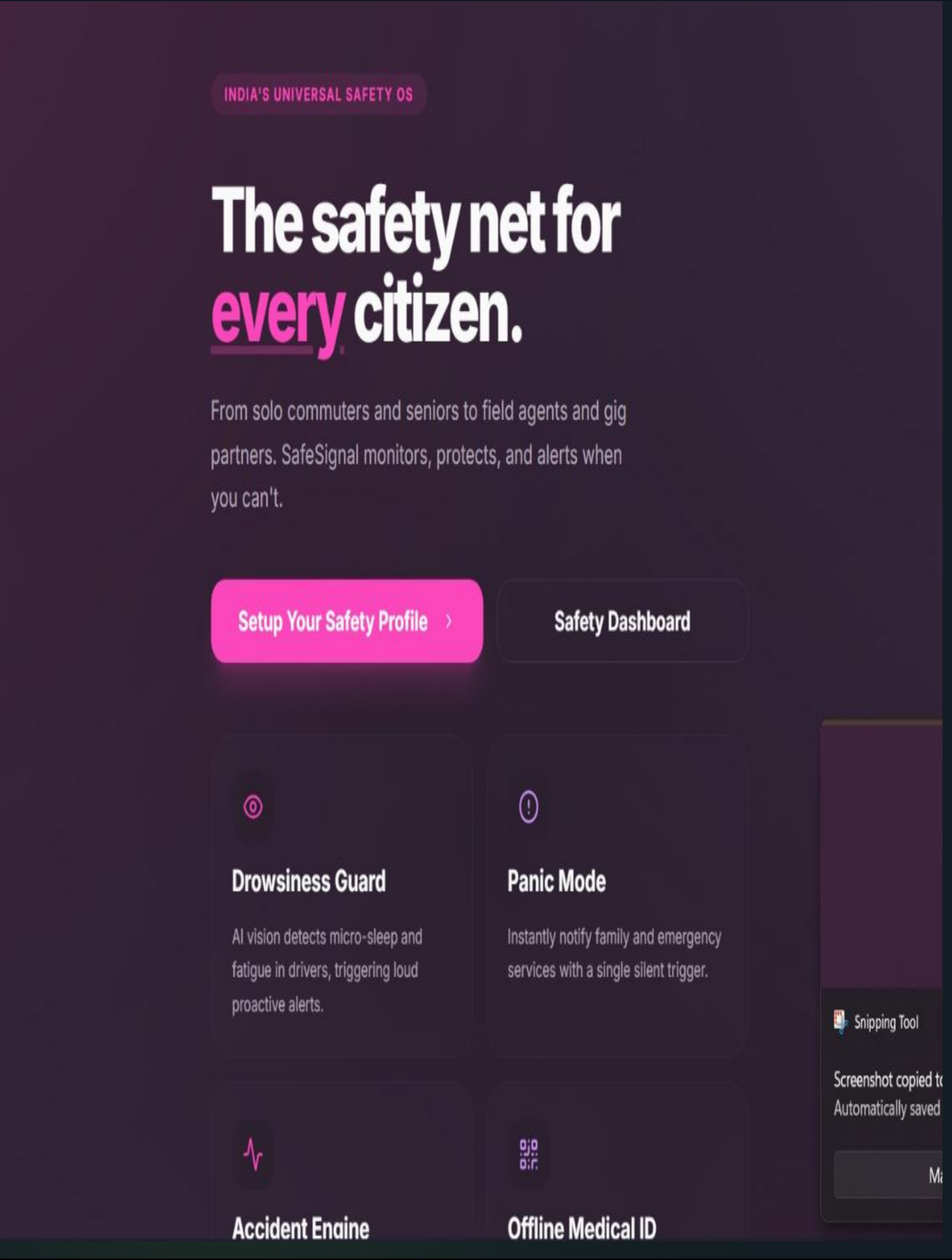

SafeSignal

An AI-Based Emergency Response System for common people .

"SAVING LIVES WHEN PLATFORMS DON'T"

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Introducing SafeSignal

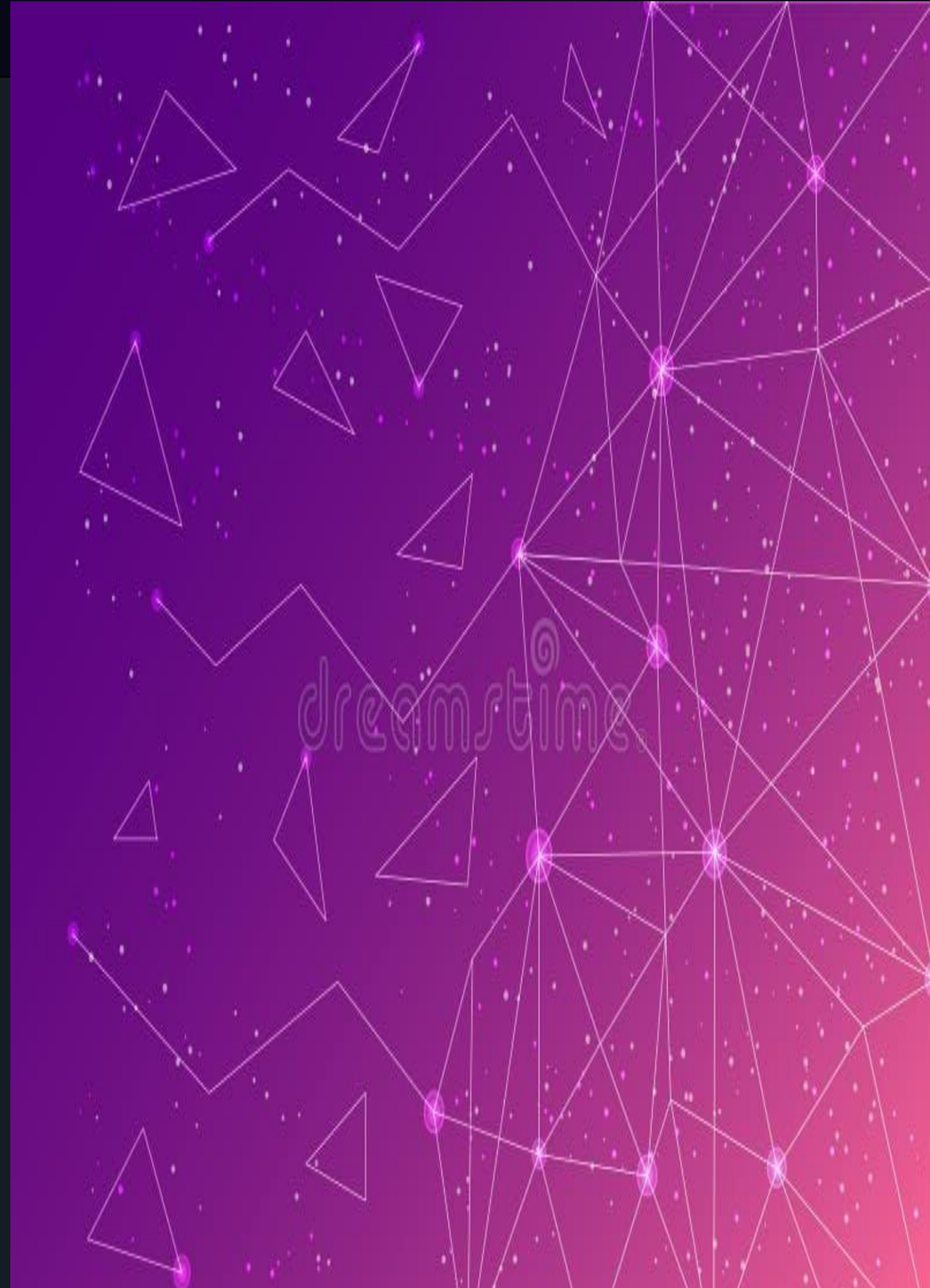
Our Vision

To build a nationwide safety net for gig workers and common people where no accident goes unnoticed and no worker is left without help.
We envision a system that bridges the gap between workers, emergency services, and platforms using AI-driven real-time response.
Our goal is to make safety an automatic, invisible layer of every gig worker's daily life — not a privilege, but a basic right.

To develop an AI-powered system that instantly detects accidents and triggers emergency response for gig workers and common people, ensuring timely medical assistance and communication, reducing fatalities and saving lives.

Core Features

- **Automated Emergency Response**
  Triggers calls, messages, and ambulance alerts within seconds if the worker is unresponsive.
- **QR-Based Medical Card (Offline)**
  Provides instant access to blood group, allergies, and emergency contacts — even without internet.
- **Live GPS Tracking & Alerts**
  Continuously shares location updates with family and responders until the incident is resolved.
- Detects real accidents in real time using accelerometer, gyroscope, and GPS with an on-device ML model.
- Comprehensive recruiter dashboards for streamlined talent acquisition



The Challenge SafeSignal is Solving

Delayed Emergency Response

When a gig worker meets with an accident, there is often no immediate way to alert anyone. Platforms like Swiggy or Zomato only detect inactivity, not emergencies. SafeSignal solves this by automatically detecting accidents and triggering alerts within seconds, ensuring help reaches during the critical golden hour.

Lack of Medical Information at Crash Sites

In most accident scenarios, first responders have no access to vital details like blood group, allergies, or existing conditions. This delay can be life-threatening. SafeSignal addresses this with an offline QR-based medical card that instantly provides essential health information to responders, even without internet access.

No Safety Accountability for Gig Workers

Gig workers operate without formal employer responsibility, meaning no structured safety or support system exists. Companies like Uber and Ola are not immediately aware of accidents. SafeSignal bridges this gap by creating a real-time notification system that connects workers, families, emergency services, and platforms into a unified safety network.

Benefits of SafeSignal

For Gigworkers

- Instant Emergency Assistance
- Increased Personal Safety & Confidence
- The QR-based medical card ensures first responders immediately know vital details.
- Immediate Notification to associates

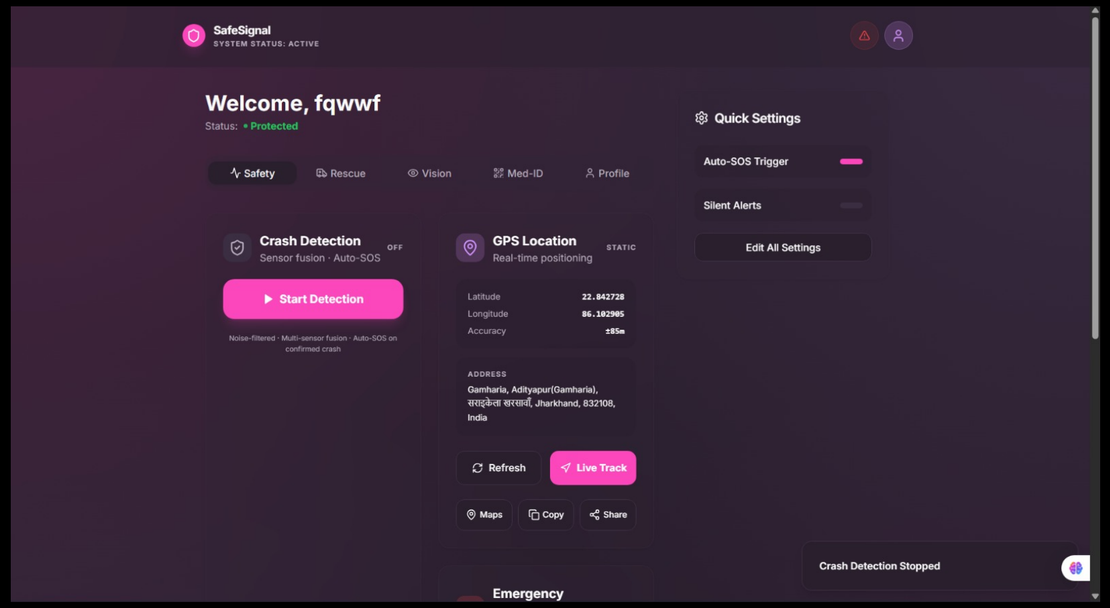
For Common People

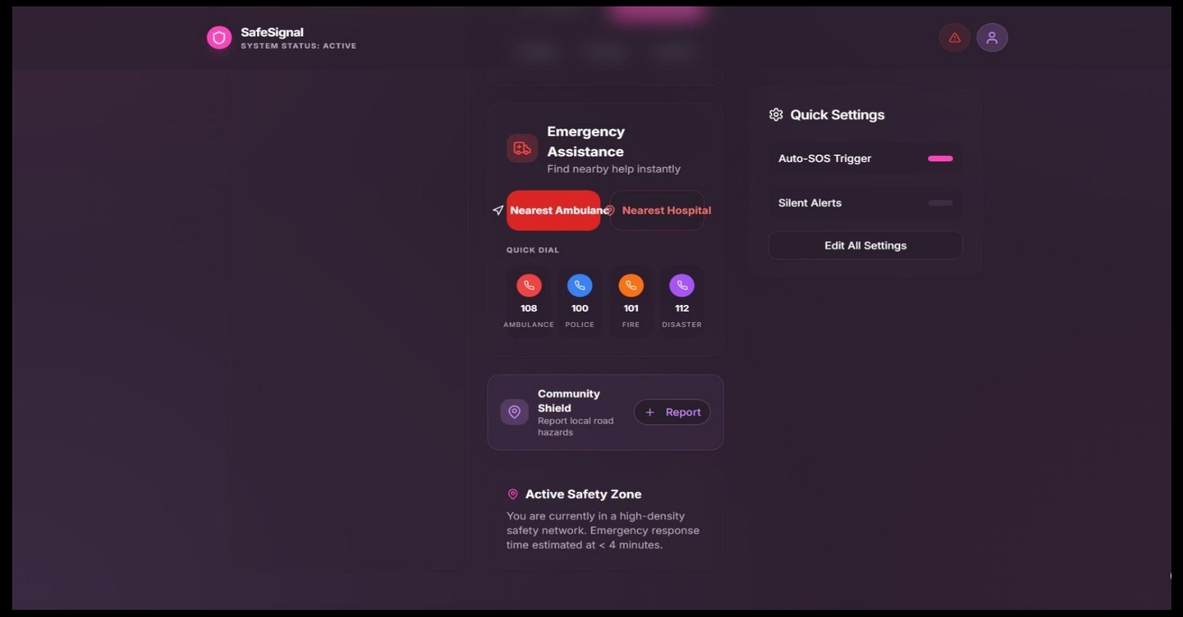
- Reduced Emergency Response Time
- Ability to Help Accident Victims Quickly
- Safer Roads for Everyone
- Increased Public Awareness & Responsibility

Additional Benefits

- Faster Ambulance Coordination
- Reduced Burden on Bystanders
- Better Urban Planning & Traffic Safety

SafeSignal: Our Implementation Progress





01

Functional Prototype

A working mobile application that simulates real-time accident detection using sensor data and triggers the complete emergency response flow. It demonstrates QR medical access, automated alerts, and live tracking in a controlled demo scenario.

02

Scalable & Secure Platform

Built on a modular cloud architecture with APIs and microservices, SafeSignal can scale seamlessly to support millions of gig workers across multiple platforms. Sensitive data is protected using strong encryption and secure authentication, ensuring privacy and reliability at every step.

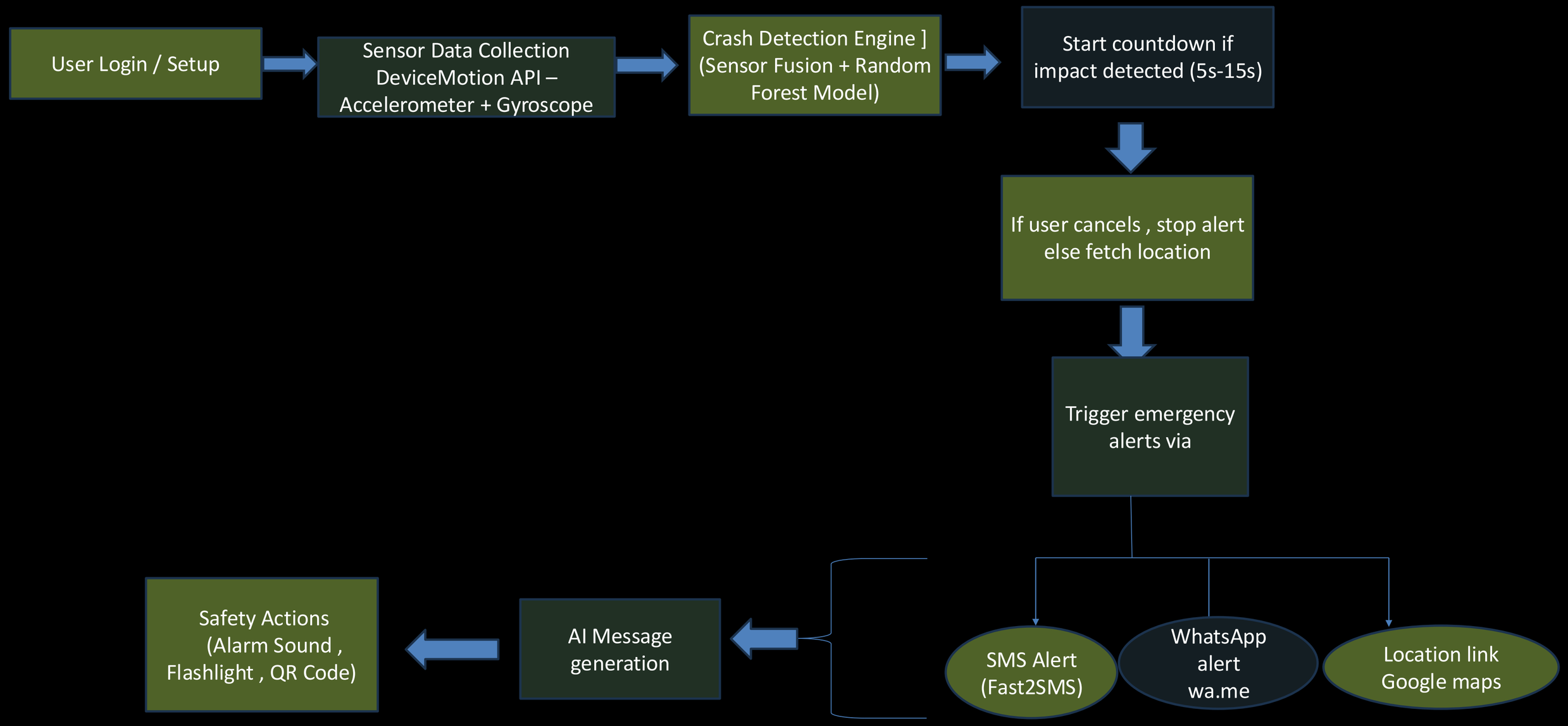
03

Next Steps & Roadmap

In the next phase, we will conduct real-world pilot testing with gig workers, refine the AI model using live data, and onboard platform partners for integration. and build predictive safety analytics to prevent accidents before they occur.

Explore our prototype:https://github.com/Spookyvegeta/guardianride

THE WORK FLOW



Technical Approach & Architecture

Tech Stack Breakdown:

Frontend

Built with **Next.js** and styled using TailwindCSS components. Integrated libraries like Recharts, qrcode.react, and Lucide React for data visualization, QR generation, and icons.

Backend & Core API

Node.js with API and firebase auth for authentication. Database powered by Cloud Firestore and External APIs integrated: OpenStreetMap Nominatim, QR Server API, and Google Maps for location, QR, and navigation features.

Database & Storage

Uses Cloud Firestore as the primary database for storing user profiles, emergency contacts, and safety logs in real time and deployed locally for now.

AI & Computer Vision

hybrid crash detection model combining signal processing and lightweight ML. It uses noise gating for preprocessing, followed by rule-based + statistical thresholding, trained on accelerometer and gyroscope patterns to distinguish real crashes from normal motion.

Additional Technologies

Kiro IDE provided by Amazon AWS is used to host the frontend of our product.

Future Scope & Development Roadmap

AI Model Expansion

The AI model can be enhanced using large-scale real-world data to improve accuracy and reduce false detections. In the future, it can evolve into a predictive system that identifies high-risk situations in advance, integrates with wearables/smart helmets, and adapts to different vehicles and road conditions for smarter, proactive safety.

Technical Infrastructure

- On-Device AI Processing
- Integrated Communication System
- TensorFlow Lite model runs directly on the phone, ensuring fast accident detection without internet dependency.

Regulatory & Authorization

Emergency Service Integration Rules
Automated calling to services like 112 must comply with telecom and emergency response regulations to avoid misuse or false alerts.

Accessibility & Monetization Evolution

- Designed for low-cost smartphones with simple UI and multilingual support, making it accessible to all
- Diversified Revenue Ecosystem which Generates income through insurance partnerships
- Ensuring equal access irrespective of economical background

Long-term Vision

- Predictive Safety Analytics
- Policy & Infrastructure Impact
- National Accident Intelligence Network

SWOT Analysis – SafeSignal

High Validation

Strengths

- AI-based real-time accident detection using on-device ML
- Works offline (QR medical card + local processing)
- Automated end-to-end emergency response system

Opportunities

- Large and growing gig workforce (15M+ in India)
- Partnerships with **Uber and Ola**
- Government integration (e-**Shram,** labour welfare)
- Expansion into wearables and insurance ecosystem

Low Capacity ← withp.batforms like Swiggy — Zomato → High Potential

Weaknesses

- Dependence on smartphone sensor accuracy
- Requires high-quality crash data for training
- Battery usage during long work sessions
- Requires active user participation (session start)

Threats

- Platform resistance due to liability concerns
- Legal/regulatory challenges
- Competition from platform-built solutions
- Data privacy and security concerns

High Risk

Revenue Streams

Insurance Partnerships
SafeSignal partners with insurance companies like Bajaj Allianz and HDFC ERGO to offer affordable micro-insurance to gig workers.
The app provides real-time accident data and verification, reducing fraud and speeding up claims — in return, SafeSignal earns a commission from each policy sold.

Union / NGO Subscriptions
Worker unions and organizations subscribe to SafeSignal to provide safety coverage for all their members at once.
Groups like Indian Federation of App-based Transport Workers can include the app as part of membership benefits, paying a bulk fee while ensuring safety for their workers.

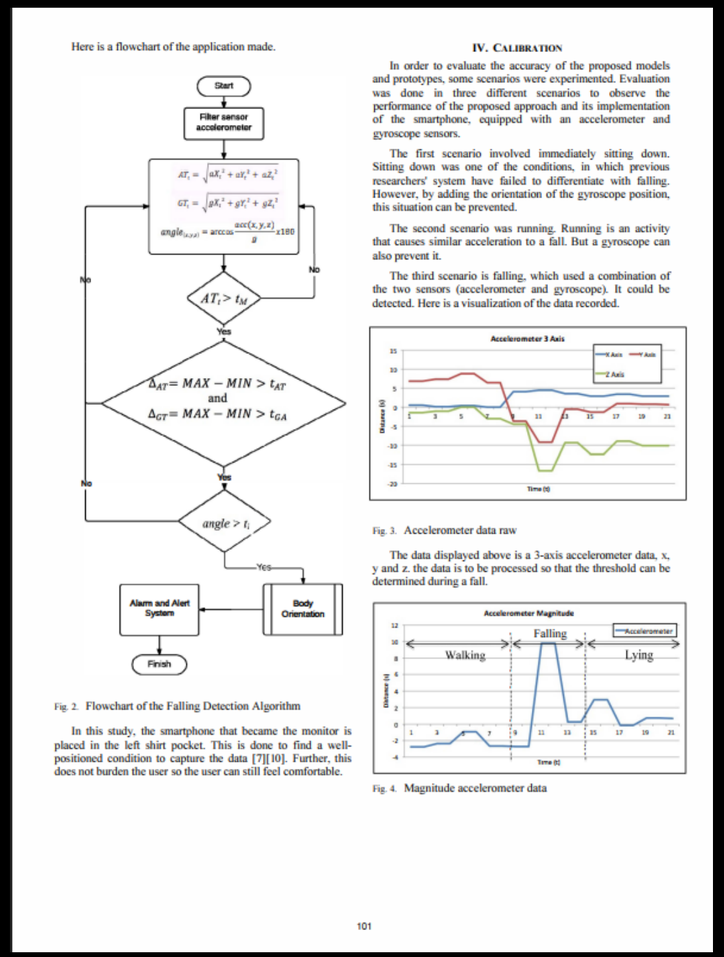
This is the most underused option for safety-focused apps. Indian companies are legally required to spend 2% of profits on CSR. Road safety is an approved CSR category. Tata Motors, Maruti, Hero MotoCorp, Bajaj — vehicle companies have enormous CSR budgets and a direct reputational interest in road safety. You approach them not as investors but as CSR sponsors: "Fund deployment of this app for 50,000 gig workers in Mumbai. We'll produce quarterly impact reports with your logo on them." No equity given. No board seats. Pure grant funding.
Your pitch to CSR team
"India has 53 road deaths every day. Your company sells the vehicles involved. A ₹25L CSR grant deploys this app to 20,000 riders and gives you measurable road safety impact data for your annual sustainability report."

FEASIBILITY

Technical Feasibility: HighThe core technology needed to detect a crash in a browser is ready, but background execution is the "final boss."Sensor Access: Browsers now have mature APIs (DeviceMotionEvent) to access high-frequency accelerometer data.On-Device AI: TensorFlow.js and WebGPU allow you to run the crash detection model locally without sending data to a server. This is highly feasible and ensures zero-latency detection.

Economic Feasibility: The "Zero-Infrastructure" AdvantageOne of the strongest points of this project is its incredibly low "Cost per User."Computing Costs ($0): Because the AI model runs on the user's phone (client-side), you don't need expensive GPU servers to process sensor data. You can scale to 1 million users without increasing your ML server bill.Database (Freemium): Using Supabase, the initial 50,000 users will likely fit within the free/low-cost tiers.Alerting Costs (Variable): Twilio/SMS APIs are the only significant cost.Risk: A buggy AI that sends "False Alarms" will drain your budget.Fix: The 30-second user-verification timer is an economic safeguard as much as a safety one.

RESEARCH FROM SOURCES



Here is a flowchart of the application made.

Fig. 2. Flowchart of the Falling Detection Algorithm

In this study, the smartphone that became the monitor is placed in the left shirt pocket. This is done to find a well-positioned condition to capture the data [7][10]. Further, this does not burden the user so the user can still feel comfortable.

IV. CALIBRATION

In order to evaluate the accuracy of the proposed models and prototypes, some scenarios were experimented. Evaluation was done in three different scenarios to observe the performance of the proposed approach and its implementation of the smartphone, equipped with an accelerometer and gyroscope sensors.

The first scenario involved immediately sitting down. Sitting down was one of the conditions, in which previous researchers' system have failed to differentiate with falling. However, by adding the orientation of the gyroscope position, this situation can be prevented.

The second scenario was running. Running is an activity that causes similar acceleration to a fall. But a gyroscope can also prevent it.

The third scenario is falling, which used a combination of the two sensors (accelerometer and gyroscope). It could be detected. Here is a visualization of the data recorded.

Fig. 3. Accelerometer data raw

The data displayed above is a 3-axis accelerometer data, x, y and z. the data is to be processed so that the threshold can be determined during a fall.

Fig. 4. Magnitude accelerometer data

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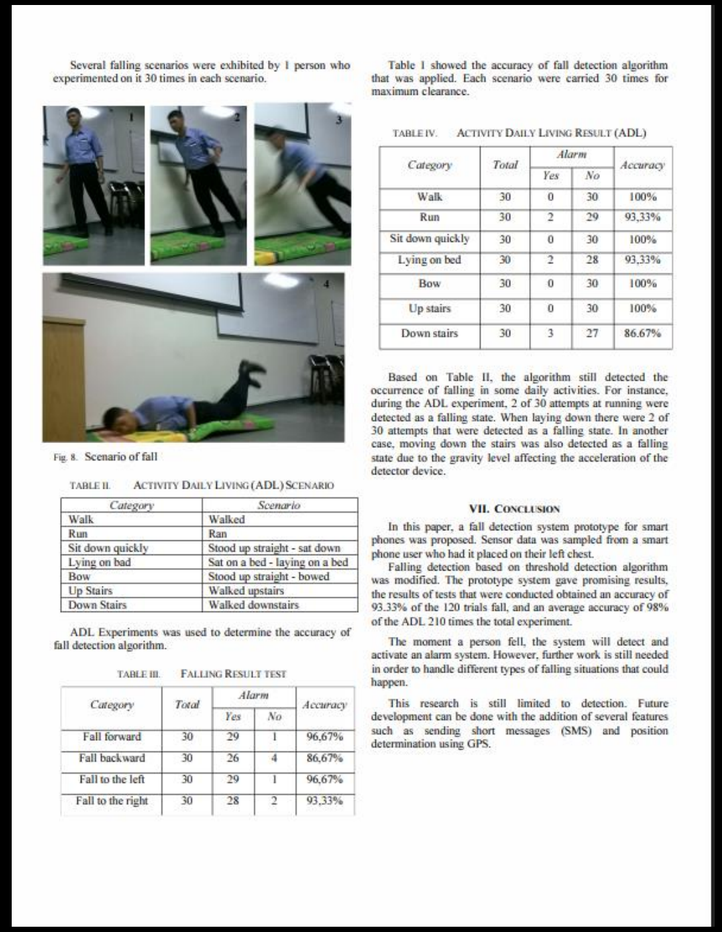


Several falling scenarios were exhibited by 1 person who experimented on it 30 times in each scenario.

Fig. 8. Scenario of fall

TABLE II. ACTIVITY DAILY LIVING (ADL) SCENARIO

Category	Scenario
Walk	Walked
Run	Ran
Sit down quickly	Stood up straight - sat down
Lying on bad	Sat on a bed - laying on a bed
Bow	Stood up straight - bowed
Up Stairs	Walked upstairs
Down Stairs	Walked downstairs

ADL Experiments was used to determine the accuracy of fall detection algorithm.

TABLE III. FALLING RESULT TEST

Category	Total	Alarm		Accuracy
		Yes	No	
Fall forward	30	29	1	96,67%
Fall backward	30	26	4	86,67%
Fall to the left	30	29	1	96,67%
Fall to the right	30	28	2	93,33%

Table 1 showed the accuracy of fall detection algorithm that was applied. Each scenario were carried 30 times for maximum clearance.

TABLE IV. ACTIVITY DAILY LIVING RESULT (ADL)

Category	Total	Alarm		Accuracy
		Yes	No	
Walk	30	0	30	100%
Run	30	2	29	93,33%
Sit down quickly	30	0	30	100%
Lying on bed	30	2	28	93,33%
Bow	30	0	30	100%
Up stairs	30	0	30	100%
Down stairs	30	3	27	86.67%

Based on Table II, the algorithm still detected the occurrence of falling in some daily activities. For instance, during the ADL experiment, 2 of 30 attempts at running were detected as a falling state. When laying down there were 2 of 30 attempts that were detected as a falling state. In another case, moving down the stairs was also detected as a falling state due to the gravity level affecting the acceleration of the detector device.

VII. CONCLUSION

In this paper, a fall detection system prototype for smart phones was proposed. Sensor data was sampled from a smart phone user who had it placed on their left chest.

Falling detection based on threshold detection algorithm was modified. The prototype system gave promising results, the results of tests that were conducted obtained an accuracy of 93.33% of the 120 trials fall, and an average accuracy of 98% of the ADL 210 times the total experiment.

The moment a person fell, the system will detect and activate an alarm system. However, further work is still needed in order to handle different types of falling situations that could happen.

This research is still limited to detection. Future development can be done with the addition of several features such as sending short messages (SMS) and position determination using GPS.

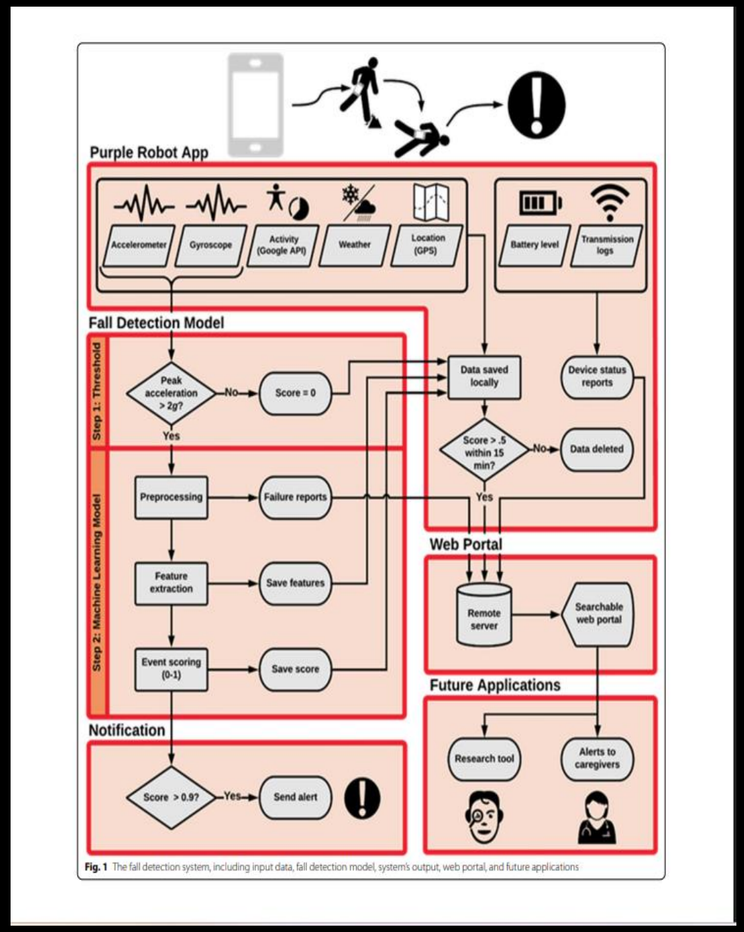


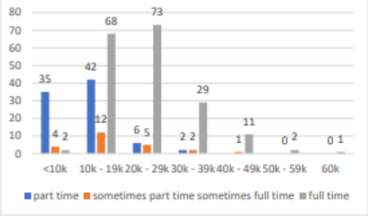
Fig. 1 The fall detection system, including input data, fall detection model, system's output, web portal, and future applications

https://www.researchgate.net/publication/279533573_Fall_Detection_System_Using_Accelerometer_and_Gyroscope_Based_on_Smartphone

https://www.researchgate.net/publication/299855987_Car_crash_detection_on_smartphones

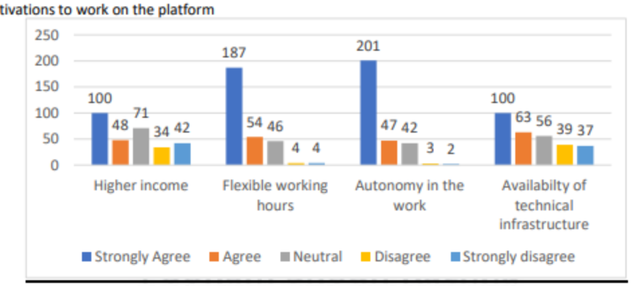
Research on the average accidents of gig workers belonging to different age groups

Most of the Uber drivers earn an income between 10k and 19k in a month.



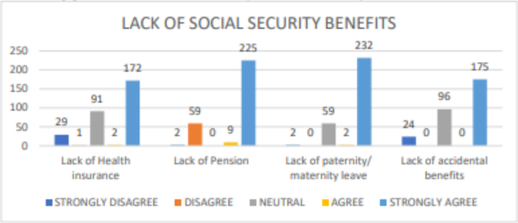
Most of the gig workers who give full time of their work in Uber earn an income in between 20k to 29k in a month and most of the gig workers who work part-time in Uber earn an income in between 10k to 19k in a month. Full time gig workers earn more than the part time gig workers.

Motivations to work on the platform



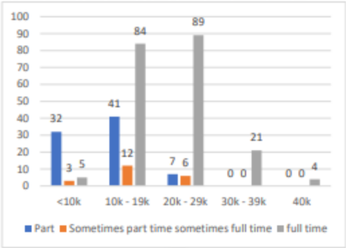
From the above table, it is found that flexible working hours and autonomy in the work are the main motivations for the gig workers to work in Uber.

Responses of the Uber gig workers on the availability of social security benefits:-



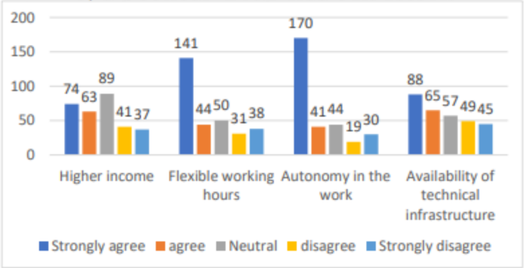
Most of them said that there is a lack of social security benefits, like a lack of health insurance, a lack of pension, a lack of paternity/ maternity leave and a lack of accidental benefits found in the platform.

Most of the Swiggy workers earn an income in between 10k-19k in a month



Most of the gig workers who give their full time to their work in Swiggy earn an income between 10k to 29k in a month. The working conditions are such that almost all are bound to work full time.

Motivations to work in the platform



- From the above table, it is found that flexible working hours and autonomy in the work are the main motivations for the gig workers to work in Swiggy.

Responses of the Swiggy gig workers on the availability of social security benefits:-

https://www.sciencedirect.com/science/article/abs/pii/S3050475926001284?utm_source=chatgpt.com

THANK YOU!